

AI & Machine Learning Fundamentals Report

Introduction

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence, such as learning, reasoning, decision-making, and problem-solving. Machine Learning (ML) is a subset of AI that allows computers to learn from data without being explicitly programmed.

What is Artificial Intelligence (AI)?

Artificial Intelligence is the simulation of human intelligence by machines. AI systems can analyze data, recognize patterns, make decisions, and improve their performance over time.

Examples

- Virtual assistants (Siri, Alexa, Google Assistant)
- Self-driving cars
- Face recognition
- Chatbots

What is Machine Learning (ML)?

Machine Learning is a branch of AI where computers learn from data and improve their predictions without explicit programming.

Examples

- Email spam detection
- Movie recommendations
- Fraud detection
- Stock price prediction

Difference Between AI and ML

Artificial Intelligence	Machine Learning
Broader field	Subset of AI
Mimics human intelligence	Learns from data
Includes reasoning and planning	Focuses on prediction and learning
Can work with or without ML	Always part of AI

Types of Artificial Intelligence